

Цель работы: выработать навык анализа сетевого трафика.

Выполнение заданий: Вариант 11

Скриншоты из Wireshark, иллюстрирующие анализ сетевого трафика.

Скриншот 1: Wireshark - Беспроводная сеть

Интерфейс Wireshark с заголовком "Беспроводная сеть". Вкладки: Файл, Редактирование, Просмотр, Запуск, Захват, Анализ, Статистика, Телефония, Беспроводной, Инструменты, Помощь. Адресная строка: http. Таблица пакетов пуста.

Скриншот 2: Wireshark - Интернет-протокол

Интерфейс Wireshark с заголовком "Интернет-протокол". Вкладки: Файл, Редактирование, Просмотр, Запуск, Захват, Анализ, Статистика, Телефония, Беспроводной, Инструменты, Помощь. Адресная строка: http || arp || icmp. Таблица пакетов содержит данные о пакетах ARP и Broadcast.

No.	Time	Source	Destination	Protocol	Length	Info
34	1.649000	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.115? Tell 192.168.0.1
285	23.357470	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.112? Tell 192.168.0.1
347	29.234035	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.119? Tell 192.168.0.1
348	29.234076	IntelCor_lc:87:bb	Tp-LinkT_bd:1a:64	ARP	42	192.168.0.119 is at 84:a6:c8:1c:87:bb
788	55.635006	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.112? Tell 192.168.0.1
797	56.739543	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.112? Tell 192.168.0.1
818	59.094536	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.119? Tell 192.168.0.1
819	59.094559	IntelCor_lc:87:bb	Tp-LinkT_bd:1a:64	ARP	42	192.168.0.119 is at 84:a6:c8:1c:87:bb
1081	67.934081	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.115? Tell 192.168.0.1
1361	89.097257	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.119? Tell 192.168.0.1
1362	89.097279	IntelCor_lc:87:bb	Tp-LinkT_bd:1a:64	ARP	42	192.168.0.119 is at 84:a6:c8:1c:87:bb
1471	98.006006	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.115? Tell 192.168.0.1
1723	119.407406	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.119? Tell 192.168.0.1
1724	119.407447	IntelCor_lc:87:bb	Tp-LinkT_bd:1a:64	ARP	42	192.168.0.119 is at 84:a6:c8:1c:87:bb
1799	123.912862	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.112? Tell 192.168.0.1
1833	128.008799	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.115? Tell 192.168.0.1
2118	149.512558	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.119? Tell 192.168.0.1
2119	149.512596	IntelCor_lc:87:bb	Tp-LinkT_bd:1a:64	ARP	42	192.168.0.119 is at 84:a6:c8:1c:87:bb
2241	155.861171	Tp-LinkT_bd:1a:64	Broadcast	ARP	42	Who has 192.168.0.112? Tell 192.168.0.1

Frame 34: 42 bytes on wire (336 bits), 42 bytes captured (336 bits) on interface \Device\NPF_{C9BC9A2F-08B4-4720-95BA-CC645058EB3A}, id 0
> Ethernet II, Src: Tp-LinkT_bd:1a:64 (d8:47:32:bd:1a:64), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> Address Resolution Protocol (request)

0000 ff ff ff ff ff ff d8 47 32 bd 1a 64 08 06 00 01G 2..d....
0010 08 00 06 04 00 01 d8 47 32 bd 1a 64 c0 a8 00 01G 2..d....
0020 00 00 00 00 00 00 c0 a8 00 73s

Интерфейс Wireshark с заголовком "Интернет-протокол". Вкладки: Файл, Редактирование, Просмотр, Запуск, Захват, Анализ, Статистика, Телефония, Беспроводной, Инструменты, Помощь. Адресная строка: http || arp || icmp. Таблица пакетов содержит данные о пакетах ARP и Broadcast.

Wireshark interface showing network traffic analysis. The top panel displays the packet list, and the bottom panel shows the packet details and hex data.

Packet List:

No.	Time	Source	Destination	Protocol	Length	Info
4032	247.002174	192.168.0.119	37.45.176.80	UDP	102	51054 → 63633 Len=60
4033	247.002200	37.45.176.80	192.168.0.119	UDP	102	63633 → 51054 Len=60
4034	247.002232	192.168.0.119	37.45.176.80	UDP	104	51054 → 63633 Len=62
4035	247.0022410	37.45.176.80	192.168.0.119	UDP	104	63633 → 51054 Len=62
4036	247.151639	192.168.0.115	192.168.0.119	TCP	74	53736 → 53561 [PSH, ACK] Seq=22993 Ack=2517 Win=30476 Len=20
4037	247.151809	192.168.0.119	192.168.0.115	TCP	74	53561 → 53736 [PSH, ACK] Seq=2537 Ack=23013 Win=30620 Len=20
4038	247.214736	192.168.0.115	192.168.0.119	TCP	74	[TCP Spurious Retransmission] 53736 → 53561 [PSH, ACK] Seq=22993 Ack=2517 Win=30476 Len=20
4039	247.214805	192.168.0.119	192.168.0.115	TCP	66	[TCP Dup ACK 4037#1] 53561 → 53736 [ACK] Seq=2557 Ack=23013 Win=30620 Len=0 SLE=22993 SRE=23013
4040	247.482382	192.168.0.115	192.168.0.119	TCP	66	[TCP Dup ACK 4031#1] 53736 → 53561 [ACK] Seq=23013 Ack=2517 Win=30476 Len=0 SLE=1785 SRE=2517
4041	247.486557	192.168.0.115	192.168.0.119	TCP	74	53736 → 53561 [PSH, ACK] Seq=23013 Ack=2537 Win=30456 Len=20
4042	247.489282	192.168.0.115	192.168.0.119	TCP	66	[TCP Dup ACK 4041#1] 53736 → 53561 [ACK] Seq=23033 Ack=2537 Win=30456 Len=0 SLE=2273 SRE=2537
4043	247.526957	192.168.0.119	192.168.0.115	TCP	54	53561 → 53736 [ACK] Seq=2557 Ack=23033 Win=30600 Len=0
4044	247.527159	192.168.0.119	37.45.176.80	UDP	102	51054 → 63633 Len=60
4045	247.569754	37.45.176.80	192.168.0.119	UDP	102	63633 → 51054 Len=60
4046	247.570004	192.168.0.119	37.45.176.80	UDP	104	51054 → 63633 Len=62
4047	247.570190	37.45.176.80	192.168.0.119	UDP	104	63633 → 51054 Len=62
4048	247.827214	192.168.0.119	192.168.0.115	TCP	74	[TCP Retransmission] 53561 → 53736 [PSH, ACK] Seq=2537 Ack=23033 Win=30600 Len=20
4049	247.861754	192.168.0.119	192.168.0.115	TCP	94	[TCP Spurious Retransmission] 53736 → 53561 [PSH, ACK] Seq=22993 Ack=2537 Win=30456 Len=40
4050	247.861824	192.168.0.119	192.168.0.115	TCP	66	[TCP Dup ACK 4043#1] 53561 → 53736 [ACK] Seq=2557 Ack=23033 Win=30600 Len=0 SLE=22993 SRE=23033

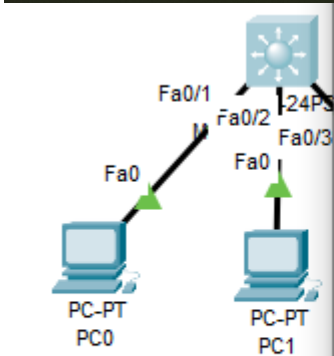
Packet Details:

- Frame 4033: 102 bytes on wire (816 bits), 102 bytes captured (816 bits) on interface \Device\NPF_{C9BC9A2F-08B4-4720-95BA-CC645058EB3A}, id 0
- Ethernet II, Src: Tp-LinkT_bd:1a:64 (d8:47:32:bd:1a:64), Dst: IntelCor_1c:87:bb (84:a6:c8:1c:87:bb)
- Internet Protocol Version 4, Src: 37.45.176.80, Dst: 192.168.0.119
- User Datagram Protocol, Src Port: 63633, Dst Port: 51054
- Data (60 bytes)

Hex Data:

```
0000 84 a6 c8 1c 87 bb d8 47 32 bd 1a 64 08 00 45 40 .....G 2..d..E@
0010 00 58 e8 bd 00 00 78 11 c2 fa 25 2d b0 50 c0 a8 ..X...x...%..P..
0020 00 77 f8 91 c7 6e 00 44 25 ef 60 0e 11 5c 00 14 ..w...D %'...\..
0030 11 80 20 01 00 00 28 4a 03 64 00 af 07 6e da d2 .....J .d.....
0040 4f af 20 01 00 00 28 4a 03 64 34 cc ea 80 d1 ca 0...J .d.....
0050 0a 74 e8 b4 c9 b4 00 14 09 ae b4 20 0b 5a 80 00 .t.....Z...
0060 b5 5e 85 ff 03 4c .....L
```

Layer 3 Switch



```
Switch>en
Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#vlan 4
Switch(config-vlan)#name VLAN4
Switch(config-vlan)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console
Switch#
```



Top

Copper Cross-Over

Toggle PDU L

Cisco Packet Tracer

File Edit Options View Tools Extensions Windows

Logical Physical x 311, y: 180

Device Name: Multilayer Switch0
Device Model: 3560-24PS
Hostname: Switch

Port	Link	VLAN	IP Address	IPv6 Address	MAC Address
FastEthernet0/1	Up	1	<not set>	<not set>	0003.E433.A801
FastEthernet0/2	Up	1	<not set>	<not set>	0003.E433.A802
FastEthernet0/3	Up	1	<not set>	<not set>	0003.E433.A803
FastEthernet0/4	Down	1	<not set>	<not set>	0003.E433.A804
FastEthernet0/5	Down	1	<not set>	<not set>	0003.E433.A805
FastEthernet0/6	Down	1	<not set>	<not set>	0003.E433.A806
FastEthernet0/7	Down	1	<not set>	<not set>	0003.E433.A807
FastEthernet0/8	Down	1	<not set>	<not set>	0003.E433.A808
FastEthernet0/9	Down	1	<not set>	<not set>	0003.E433.A809
FastEthernet0/10	Down	1	<not set>	<not set>	0003.E433.A80A
FastEthernet0/11	Down	1	<not set>	<not set>	0003.E433.A80B
FastEthernet0/12	Down	1	<not set>	<not set>	0003.E433.A80C
FastEthernet0/13	Down	1	<not set>	<not set>	0003.E433.A80D
FastEthernet0/14	Down	1	<not set>	<not set>	0003.E433.A80E
FastEthernet0/15	Down	1	<not set>	<not set>	0003.E433.A80F
FastEthernet0/16	Down	1	<not set>	<not set>	0003.E433.A810
FastEthernet0/17	Down	1	<not set>	<not set>	0003.E433.A811
FastEthernet0/18	Down	1	<not set>	<not set>	0003.E433.A812
FastEthernet0/19	Down	1	<not set>	<not set>	0003.E433.A813
FastEthernet0/20	Down	1	<not set>	<not set>	0003.E433.A814
FastEthernet0/21	Down	1	<not set>	<not set>	0003.E433.A815
FastEthernet0/22	Down	1	<not set>	<not set>	0003.E433.A816
FastEthernet0/23	Down	1	<not set>	<not set>	0003.E433.A817
FastEthernet0/24	Down	1	<not set>	<not set>	0003.E433.A818
GigabitEthernet0/1	Down	1	<not set>	<not set>	0003.E433.A819
GigabitEthernet0/2	Down	1	<not set>	<not set>	0003.E433.A81A
Vlan1	Down	1	<not set>	<not set>	00D0.FFB5.79B8

Physical Location: Intercity > Home City > Corporate Office > Main Wiring Closet > Rack > Multilayer Switch0

Time: 00:07:10

Scenario 0

New Delete

Toggle PDU List Window

Copper Cross-Over

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 4
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Top

```
!  
interface FastEthernet0/1  
  switchport access vlan 2  
  switchport mode access  
  switchport nonegotiate  
!  
interface FastEthernet0/2  
  switchport access vlan 3  
  switchport mode access  
  switchport nonegotiate  
!  
interface FastEthernet0/3  
  switchport access vlan 4  
  switchport mode access  
  switchport nonegotiate  
!  
interface FastEthernet0/4  
!  
interface FastEthernet0/5  
!  
interface FastEthernet0/6  
--More--
```

☐ Top



```
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#int vlan 2  
Switch(config-if)#  
%LINK-5-CHANGED: Interface Vlan2, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up  
  
Switch(config-if)#ip address 2.2.2.1 255.255.255.0  
Switch(config-if)#exit  
Switch(config)#int vlan 3  
Switch(config-if)#  
%LINK-5-CHANGED: Interface Vlan3, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3, changed state to up  
  
Switch(config-if)#ip address 3.3.3.1 255.255.255.0  
Switch(config-if)#exit  
Switch(config)#int vlan 4  
Switch(config-if)#  
%LINK-5-CHANGED: Interface Vlan4, changed state to up  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan4, changed state to up  
  
Switch(config-if)#ip address 4.4.4.1 255.255.255.0  
Switch(config-if)#exit  
Switch(config)#end  
Switch#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Switch#
```

☐ Top



```
!  
interface Vlan1  
  no ip address  
  shutdown  
!  
interface Vlan2  
  mac-address 00d0.ffb5.7901  
  ip address 2.2.2.1 255.255.255.0  
!  
interface Vlan3  
  mac-address 00d0.ffb5.7902  
  ip address 3.3.3.1 255.255.255.0  
!  
interface Vlan4  
  mac-address 00d0.ffb5.7903  
  ip address 4.4.4.1 255.255.255.0  
!  
ip classless  
!  
ip flow-export version 9  
!  
!  
!  
!  
!  
!  
!  
line con 0  
!  
--More-- |
```

☐ Top



Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 2.2.2.1

Pinging 2.2.2.1 with 32 bytes of data:

Reply from 2.2.2.1: bytes=32 time<1ms TTL=255

Reply from 2.2.2.1: bytes=32 time<1ms TTL=255

Reply from 2.2.2.1: bytes=32 time<1ms TTL=255

Reply from 2.2.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 2.2.2.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>

☐ Top

Copper Cross-Over

Toggle



Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 3.3.3.1

Pinging 3.3.3.1 with 32 bytes of data:

Reply from 3.3.3.1: bytes=32 time<1ms TTL=255

Reply from 3.3.3.1: bytes=32 time<1ms TTL=255

Reply from 3.3.3.1: bytes=32 time<1ms TTL=255

Reply from 3.3.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 3.3.3.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

☐ Top

Copper Cross-Over

Toggle



Command Prompt

Cisco Packet Tracer PC Command Line 1.0

C:\>ping 4.4.4.1

Pinging 4.4.4.1 with 32 bytes of data:

Reply from 4.4.4.1: bytes=32 time<1ms TTL=255

Reply from 4.4.4.1: bytes=32 time<1ms TTL=255

Reply from 4.4.4.1: bytes=32 time<1ms TTL=255

Reply from 4.4.4.1: bytes=32 time<1ms TTL=255

Ping statistics for 4.4.4.1:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>|

☐ Top

Copper Cross-Over

Toggl



```
C:\>ping 2.2.2.2
```

```
Pinging 2.2.2.2 with 32 bytes of data:
```

```
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
```

```
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
```

```
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
```

```
Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
```

```
Ping statistics for 2.2.2.2:
```

```
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
```

```
Approximate round trip times in milli-seconds:
```

```
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>
```

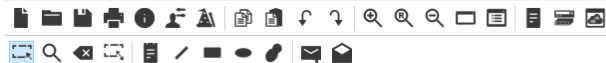
☐ Top

Copper Cross-Over

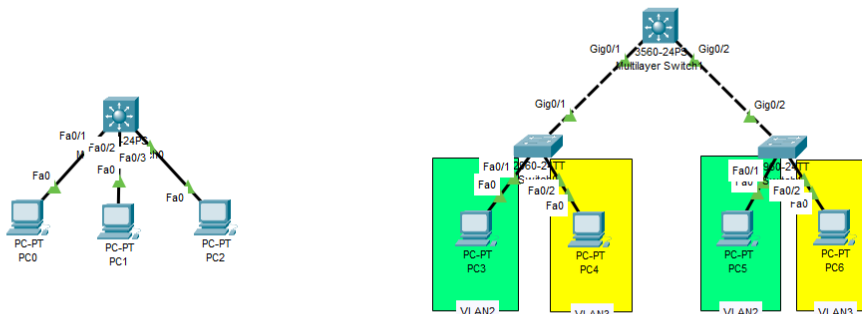
Toggle PD

Cisco Packet Tracer - C:\Users\minsk\Desktop\Work\4-й курс\Компьютерные сети\ЛР\ЛР 13.pkt

File Edit Options View Tools Extensions Window Help



Logical Physical x: 508, y: 135 Root 13:26:30



Time: 00:26:49 Realtime Simulation

Scenario 0

New Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
------	-------------	--------	-------------	------	-------	-----------	----------	-----	------

```
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#
```

☐ Top

Copper Straight-Through

Toggle PDU

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#
```

☐ Top

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3,  
Switch#conf t  
Enter configuration commands, one per line. End with CNTL/Z.  
Switch(config)#int fa0/2  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 2  
Switch(config-if)#exit  
Switch(config)#int fa0/3  
Switch(config-if)#switchport mode access  
Switch(config-if)#switchport access vlan 3  
% Access VLAN does not exist. Creating vlan 3  
Switch(config-if)#exit  
Switch(config)#int gi0/1  
Switch(config-if)#swi  
Switch(config-if)#switchport mode trunk  
  
Switch(config-if)#  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0  
  
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0  
  
Switch(config-if)#switchport mode trunk allowed vlan 2,3  
^  
% Invalid input detected at '^' marker.  
  
Switch(config-if)#switchport trunk allowed vlan 2,3  
Switch(config-if)#end  
Switch#  
%SYS-5-CONFIG_I: Configured from console by console  
  
Switch#wr mem  
Building configuration...  
[OK]  
Switch#
```

☐ Top

Copper Straight-Through

Toggle PDU L



```
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to up
%LINK-5-CHANGED: Interface FastEthernet0/3, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/3, changed state to up

Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#swi
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 2
% Access VLAN does not exist. Creating vlan 2
Switch(config-if)#exit
Switch(config)#int fa0/3
Switch(config-if)#switchport mode access
Switch(config-if)#switchport access vlan 3
% Access VLAN does not exist. Creating vlan 3
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#switchport mode trunk

Switch(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to down
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/2, changed state to up

Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

☐ Top

Press RETURN to get started!

```
Switch>en
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#int gi0/1
Switch(config-if)#swi
Switch(config-if)#switchport mode trunk
Command rejected: An interface whose trunk encapsulation is "Auto" can not be configured to "trunk" mode.
Switch(config-if)#switchport trunk enca
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#int gi0/2
Switch(config-if)#swi
Switch(config-if)#switchport trunk encapsulation dot1q
Switch(config-if)#switchport mode trunk
Switch(config-if)#switchport trunk allowed vlan 2,3
Switch(config-if)#exit
Switch(config)#int vlan 2
Switch(config-if)#ip address 2.2.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#int vlan 3
Switch(config-if)#ip address 3.3.3.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#end
Switch#
%SYS-S-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

[Top](#)



Physical Config CLI Attributes

```
Switch(config-if)#ip address 2.2.2.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#int vlan 3
Switch(config-if)#ip address 3.3.3.1 255.255.255.0
Switch(config-if)#exit
Switch(config)#ip routing
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Switch(config)#vlan 2
Switch(config-vlan)#
%LINK-5-CHANGED: Interface Vlan2, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan2, changed state to up

Switch(config-vlan)#name VLAN2
Switch(config-vlan)#exit
Switch(config)#vlan 3
Switch(config-vlan)#
%LINK-5-CHANGED: Interface Vlan3, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Vlan3, changed state to up

Switch(config-vlan)#name VLAN3
Switch(config-vlan)#exit
Switch(config)#end
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#wr mem
Building configuration...
[OK]
Switch#
```

☐ Top



PC3

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2.2.2.1

Pinging 2.2.2.1 with 32 bytes of data:

Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time=1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time=1ms TTL=255

Ping statistics for 2.2.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

☐ Top

Copper Straight-Through

Tog

PC4

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3.3.3.1

Pinging 3.3.3.1 with 32 bytes of data:

Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time=1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255

Ping statistics for 3.3.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

☐ Top

Copper Straight-Through



PC5

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 2.2.2.1

Pinging 2.2.2.1 with 32 bytes of data:

Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255
Reply from 2.2.2.1: bytes=32 time<1ms TTL=255

Ping statistics for 2.2.2.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

☐ Top

Copper Straight-Through

Toggle



PC6

Physical Config Desktop Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 3.3.3.1

Pinging 3.3.3.1 with 32 bytes of data:

Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time<1ms TTL=255
Reply from 3.3.3.1: bytes=32 time=1ms TTL=255
Reply from 3.3.3.1: bytes=32 time=1ms TTL=255

Ping statistics for 3.3.3.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>|
```

☐ Top

Copper Straight-Through

Tog

PC6

Physical Config Desktop Programming Attributes

Command Prompt

```
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>
C:\>ping 3.3.3.2

Pinging 3.3.3.2 with 32 bytes of data:

Reply from 3.3.3.2: bytes=32 time<1ms TTL=128
Reply from 3.3.3.2: bytes=32 time<1ms TTL=128
Reply from 3.3.3.2: bytes=32 time<1ms TTL=128
Reply from 3.3.3.2: bytes=32 time=1ms TTL=128

Ping statistics for 3.3.3.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 2.2.2.2

Pinging 2.2.2.2 with 32 bytes of data:

Reply from 2.2.2.2: bytes=32 time<1ms TTL=127
Reply from 2.2.2.2: bytes=32 time=10ms TTL=127
Reply from 2.2.2.2: bytes=32 time=10ms TTL=127
Reply from 2.2.2.2: bytes=32 time=1ms TTL=127

Ping statistics for 2.2.2.2:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 10ms, Average = 5ms

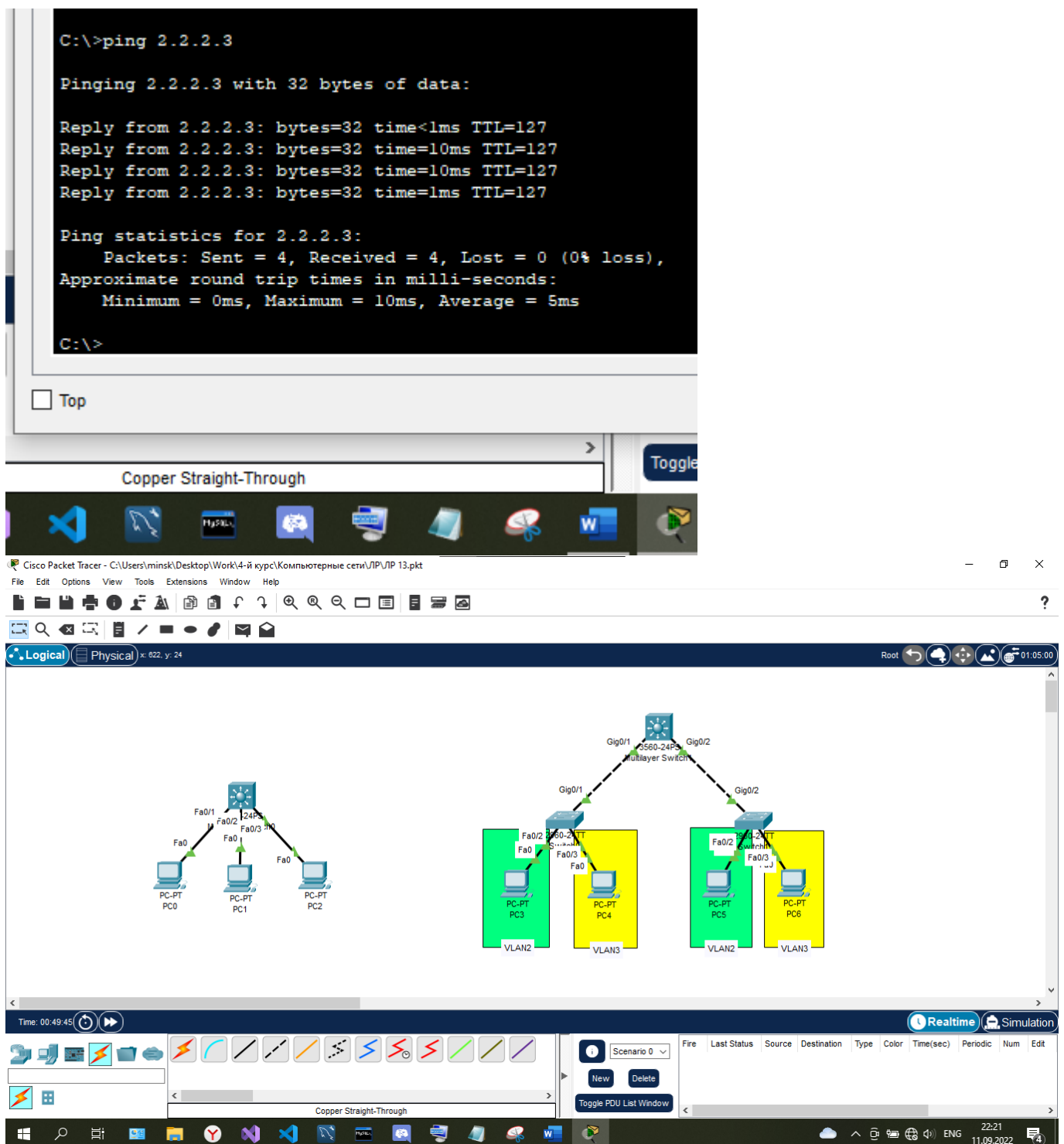
C:\>
```

☐ Top

Copper Straight-Through

Toggle PDU L





Контрольные вопросы

1. Перечислите операторы фильтров и то, что они значат

==/eq - Равно

!=/ne - Не равно

</lt - Меньше чем

<=/le - Меньше или равно

>/gt - Больше чем

>=/ge - Больше или равно

2. Как проходит Фильтрация IPv6 протокола?

ipv6 – показывает трафик IPv6

3. Как фильтровать по номеру порта?

ftp.port==<значение>

Вывод: я научился анализировать сетевой трафик.